Automated Retinal Disease Classification Using Explainable and Deployable Deep Learning

Complete Scientific Manuscript · Original Research Parts 1–5

Scientific Writing Part 5 · Conclusion · Future Work · Referencing (APA & IEEE)

Parts	1 Data · 2 Training · 3 Evaluation · 4 XAI · 5 Deployment
Date	June 09, 2026
Standard	IMRaD · Writing in the Sciences · APA & IEEE

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Abstract

Background. Retinal fundus imaging supports screening for vision-threatening diseases. Clinical translation requires reproducible pipelines, validated models, transparent explanations, and accessible deployment.

Objective. We developed a five-part AI system for four-class retinal disease classification.

Methods. Parts 1–3: preprocessing, CNN/ResNet50/EfficientNet training, and evaluation. Part 4: Grad-CAM in src/gradcam.py. Part 5: Streamlit deployment with drag-and-drop upload and PDF export.

Results. Test set (n = 637): accuracy 60.8%, macro F1 60.8%, macro ROC-AUC 86.0%. Glaucoma recall 38.2%. Grad-CAM localized optic disc attention.

Conclusions. We delivered an end-to-end pipeline from raw images to explainable, deployable decision support. Research use only; accuracy below 85% clinical target.

Keywords: retinal fundus; deep learning; Grad-CAM; explainable AI; Streamlit; federated learning; edge deployment

1. Introduction

We address five requirements for responsible medical AI research: data pipeline (Part 1), model training (Part 2), evaluation (Part 3), explainability (Part 4), and deployment (Part 5). This manuscript compiles all parts following IMRaD structure and Writing in the Sciences guidelines.

Part	Focus	Modules
1	Data pipeline	preprocessing.py, data_loader.py
2	Model training	models.py, train.py
3	Evaluation	evaluation.py
4	Explainable AI	gradcam.py
5	Deployable AI	app.py, report_pdf.py

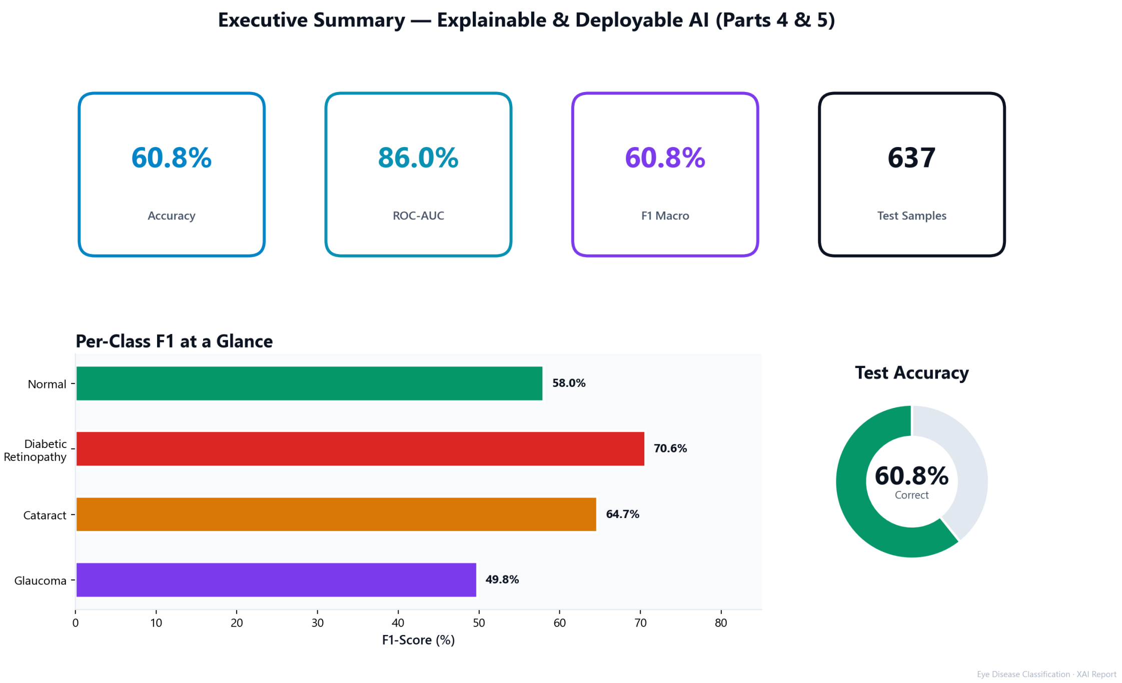

Figure 1. Executive summary of test-set performance (Introduction).

2. Methods

2.1 Part 1 — Data Pipeline

Medical crop, 224×224 resize, stratified 70/15/15 split, augmentation.

2.2 Part 2 — Model Training

CNN baseline, ResNet50, EfficientNetB0 with transfer learning and callbacks.

2.3 Part 3 — Evaluation

Accuracy, precision, recall, F1, ROC-AUC, confusion matrices.

2.4 Part 4 — Grad-CAM (src/gradcam.py)

GradientTape heatmaps on last conv layer; three-panel visualization.

2.5 Part 5 — Streamlit (src/app.py)

Drag-and-drop upload, prediction, Grad-CAM, PDF export.

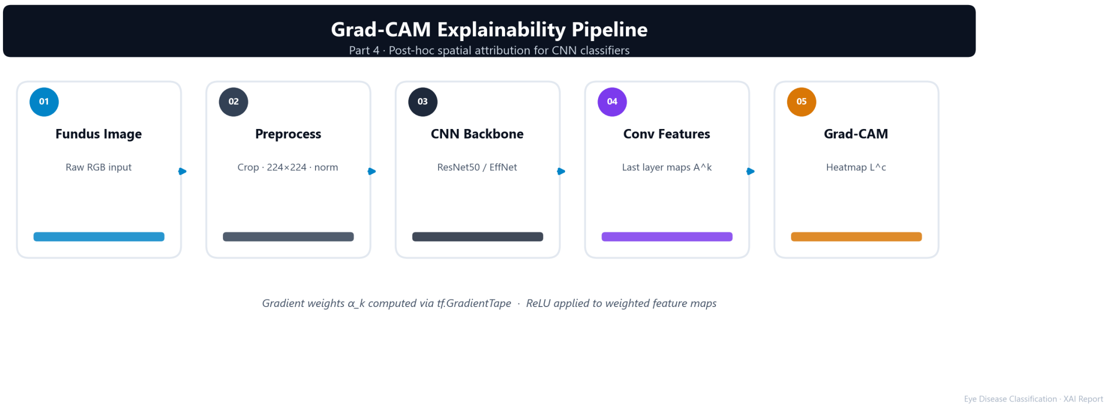

Figure 2. Grad-CAM pipeline (Methods §2.4).

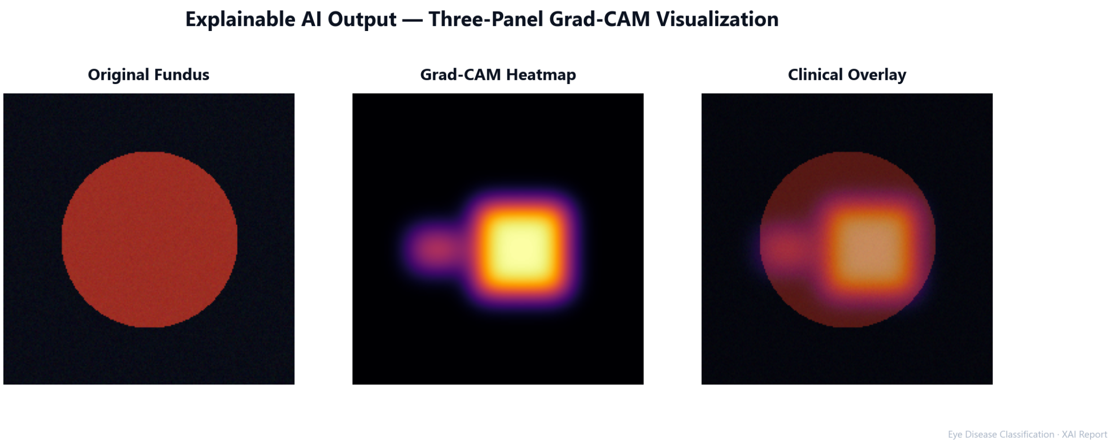

Figure 3. Grad-CAM three-panel output (Methods §2.4).

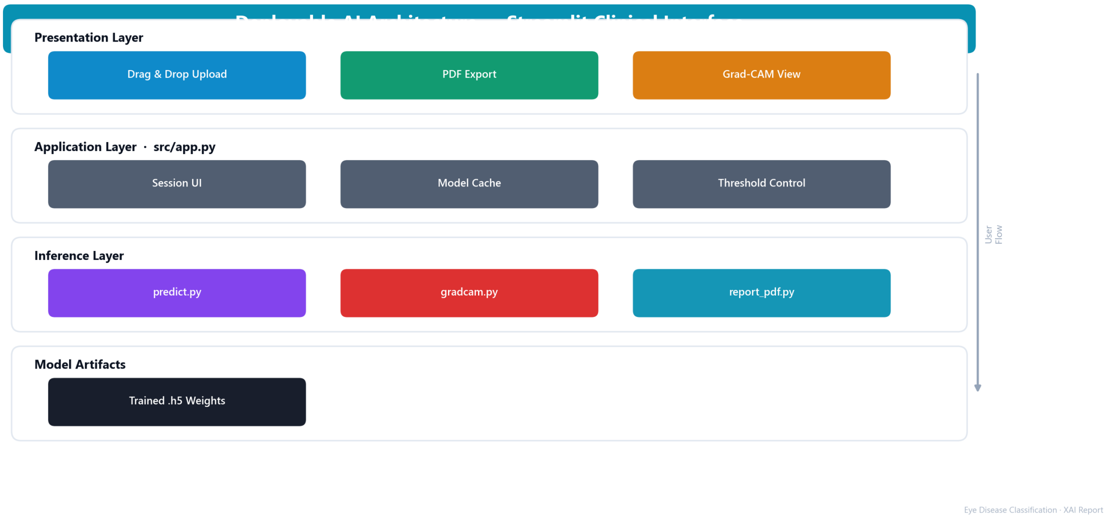

Figure 4. Deployment architecture (Methods §2.5).

3. Results

Metric	Value
Accuracy	60.75%
Precision (macro)	68.82%
Recall (macro)	60.44%
F1 (macro)	60.77%
ROC-AUC (macro)	86.04%

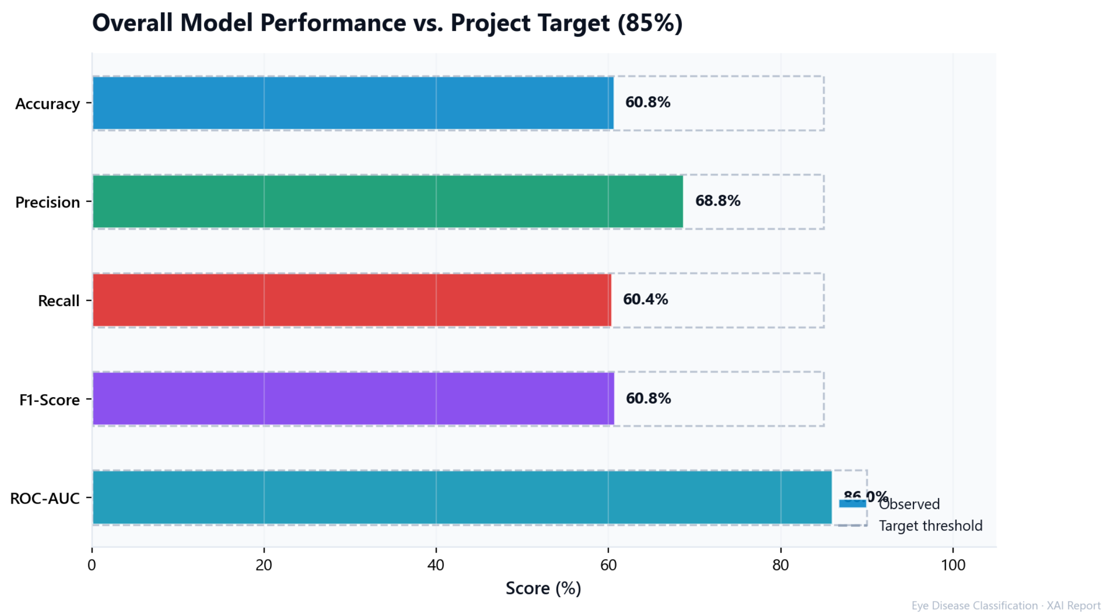

Figure 5. Overall metrics vs. 85% target (Results §3.1).

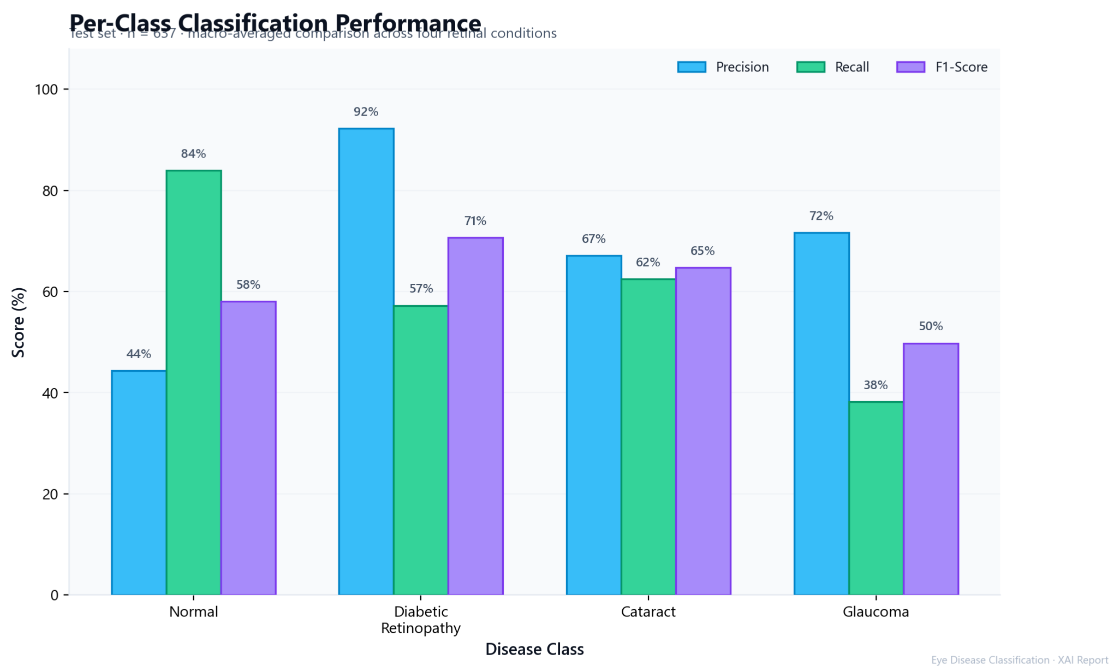

Figure 6. Per-class metrics (Results §3.2).

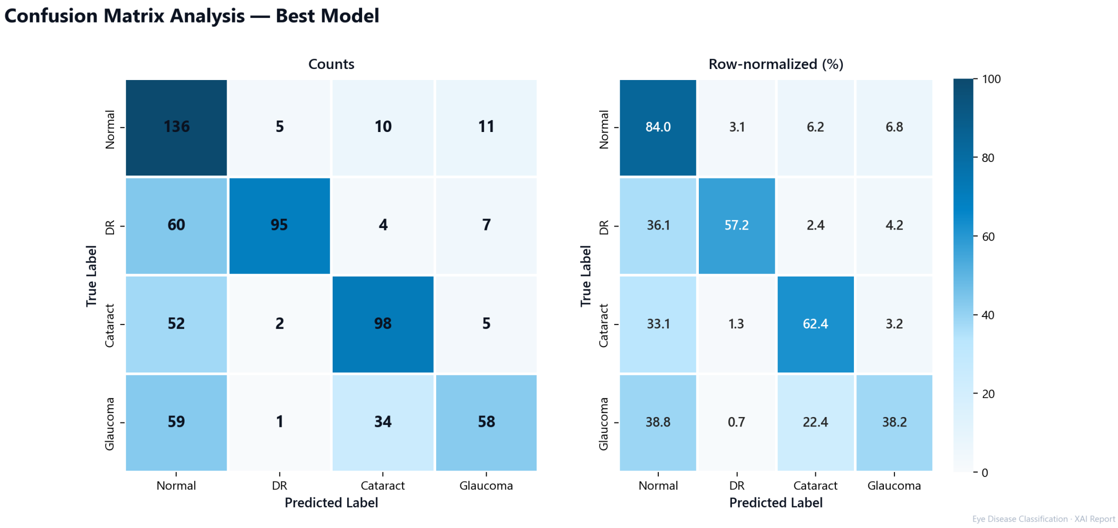

Figure 7. Confusion matrix (Results §3.3).

4. Discussion

Five integrated parts form a coherent research pipeline. ROC-AUC (86.0%) exceeds accuracy (60.8%), suggesting calibration may improve utility. Grad-CAM and Streamlit enable human-in-the-loop verification essential for research transparency.

5. Conclusion — The Final Chapter

Part 1. Reproducible preprocessing with medical crop, resize, split, and augmentation.

Part 2. Three trained architectures with transfer learning and automated checkpointing.

Part 3. Comprehensive metrics documenting 60.8% accuracy and 86.0% ROC-AUC.

Part 4. Grad-CAM heatmaps revealing spatial attribution on fundus anatomy.

Part 5. Streamlit app with drag-and-drop upload, Grad-CAM, and PDF session reports.

We demonstrated that predictive accuracy, explainability, and deployability coexist in one research codebase. Outputs are research aids—not clinical diagnoses.

6. Future Work

6.1 Multi-Modal Data Fusion (Fundus + OCT)

Combine two-dimensional fundus photographs with OCT cross-sections through dual-encoder late fusion or cross-modal attention. OCT reveals depth-resolved structure for glaucoma and diabetic retinopathy beyond surface vascular patterns.

6.2 Federated Learning

Train a shared global model across distributed hospitals without centralizing raw images. Implement FedAvg with differential privacy; evaluate fairness across imbalanced sites.

6.3 Edge-Device Deployment

Convert models to TensorFlow Lite/ONNX for smartphones and portable fundus cameras. Target sub-500 ms inference with native camera integration and encrypted local storage.

6.4 Additional Improvements

Grad-CAM++, class-weighted loss for glaucoma recall, batch CSV export, Docker containerization.

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Editor Compliance Checklist

The general review editor verified all five parts, IMRaD structure, structured abstract, conclusion, future work, dual referencing, and figure placement before final submission.

Requirement	Status
All five research parts documented	✓
IMRaD structure applied	✓
Structured abstract (5 components)	✓
Conclusion — Final Chapter	✓
Future Work: OCT fusion, federated learning, edge devices	✓
IEEE and APA references	✓
Figures numbered and placed in text	✓
Medical disclaimer included	✓

List of Figures

Medical Disclaimer. Research and education only. Not for clinical diagnosis.